

CHAPTER 16

Excitons Under the Influence of External Fields

In this chapter we discuss the behaviors of excitons under the influence of external fields such as magnetic and electric field.

16.1 Magnetic Fields

- Magneto-optics, i.e., the influence of a magnetic field B on excitons.
- Two “natural” energy units: the excitonic Rydberg energy Ry^* , the cyclotron energy.

The weak field limit: $Ry^* \gg \hbar\omega_c \Rightarrow \gamma = \hbar \frac{eB}{\mu Ry^*} \ll 1$

The strong-field limit: $Ry^* = \hbar\omega_c \Rightarrow \gamma \approx 1$

- A sufficiently strong magnetic field also produces quasi one-dimensional subbands:

The carriers can move freely only parallel to B ;

In the plane perpendicular to B the motion is quantized to circles in the classical picture, Landau levels:

$$E(n_l, \mathbf{k}_p) - E_0 = \left(n_l + \frac{1}{2}\right)\hbar\omega_c + \frac{\hbar^2 \mathbf{k}_p^2}{2m_{eff}}$$

The selection rules for inter-Landau level transition are within one fan of Landau-levels, parabolic bands $\forall n_l = \pm 1$, and for interband transitions $n_{l,CB} - n_{l,VB} = 0$

- A magnetic field to an exciton: the relative motion is deformed by Lorentz forces. In perturbation theory, this deformation can be described by a weak admixture of other states. And resulting the angular momentum is proportional to $\mathbf{B}$.

- Diamagnetic shift ΔE_{dia} to higher energies: $\Delta E_{dia} = aB^2$
- Exciton states that $B=0$ can be aligned relative to $\mathbf{B}$, exhibit in addition the linear Zeeman splitting.
- This magnetic moment can come from the spins of electron and hole.
- For singlet and triplet excitons with S envelope the difference or the sum of electron and hole g-factors enters: $\Delta E_z = \pm \frac{1}{2} |g_e + g_h| \mu_B B$ (16.2b)
- If the states are split for $\mathbf{B}=0$ by an amount Δ , the Zeeman splitting is suppressed until ΔE_z comparable to Δ . By solving a 2×2 Hamiltonian :

$$\begin{pmatrix} E_0 + aB^2 + \frac{\hbar^2 \mathbf{k}_p^2}{2m} - E & -\frac{1}{2} \mu_B g B \\ -\frac{1}{2} \mu_B g B & E_0 + \Delta + aB^2 + \frac{\hbar^2 \mathbf{k}_p^2}{2m} - E \end{pmatrix} \quad (16.3)$$

- For a quantitative calculation, perturbation theory for (almost) degenerate states has to be used.

16.1.1 Nonmagnetic Bulk Semiconductors

- Figure 16.1 The splitting is obviously connected with the Zeeman effect while the diamagnetic shift is hardly visible.(CdS)
- Fig. 16.2 The transmission spectra of ZnO (the $n_B = 1$ A- and B-exciton resonances for the polarization $\mathbf{E} \parallel \mathbf{c}$). The Zeeman splitting of the $A\Gamma_{1,2}$ triplet states with increasing $\mathbf{B}$.
- Figure 16.3 The splitting pattern in CdS ($n_B = 1$ A- and B-exciton resonances for the polarization $\mathbf{B} \perp \mathbf{c}$)

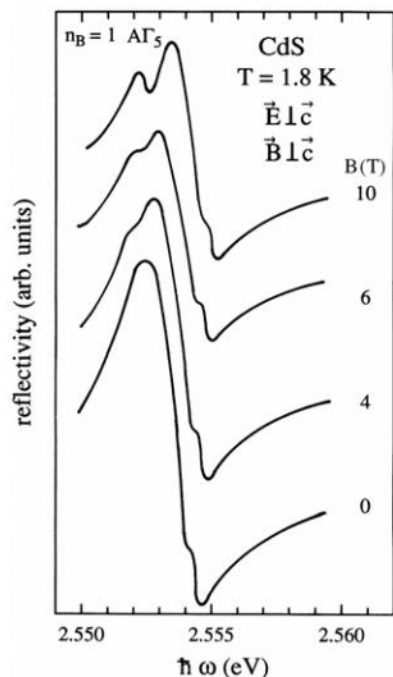


Fig. 16.1. Reflection spectra of the $n_B = 1A\Gamma_5$ -exciton resonance for various magnetic fields [82B1]

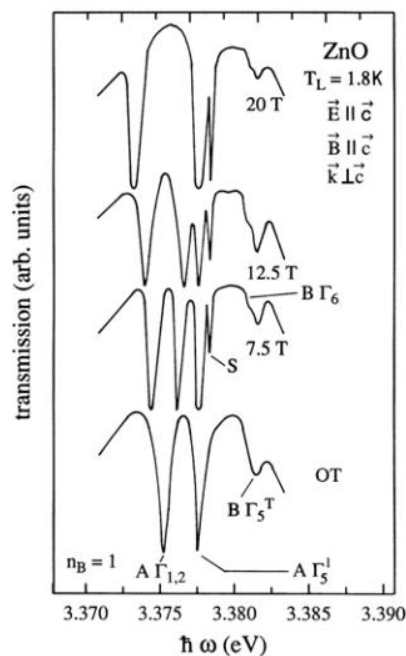


Fig. 16.2. Transmission spectra in the region of the $n_B = 1A$ and B exciton resonances of ZnO for the orientation $\mathbf{E} \parallel \mathbf{c}$ [82B1]

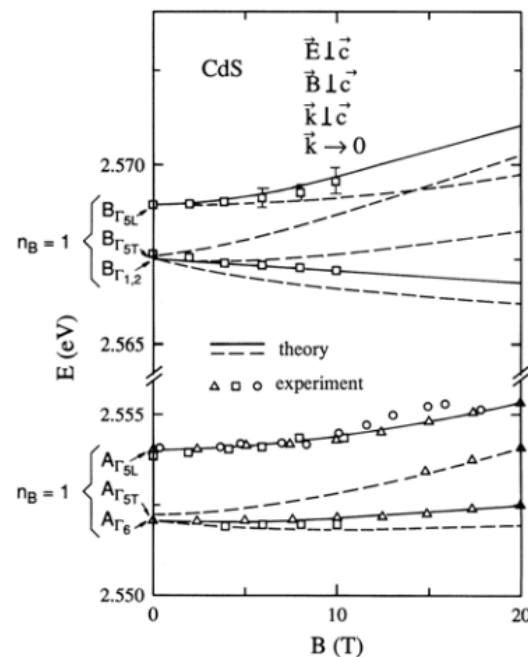


Fig. 16.3. The splitting pattern of the $n_B = 1A$ - and B-exciton resonances in CdS with increasing magnetic field [82B1,85H1]

- Figure 16.4 Dispersion relation of the $n_B = 1$ exciton polariton resonances in ZnTe.
- Fig. 16.5 The magnetic field behavior of the $n_B = 2$, $\ell = 1$ exciton resonances in ZnSe.

For higher fields this splitting increases due to Zeeman terms.

There is a pronounced diamagnetic shift due to the large radius.

- The observation of Landau levels might be difficult, but a way by investigating the continuum states.

To observe the Landau levels, their broadening must be smaller than the cyclotron energy as in micro wave experiments: $\omega_c T_2 \geq 1$

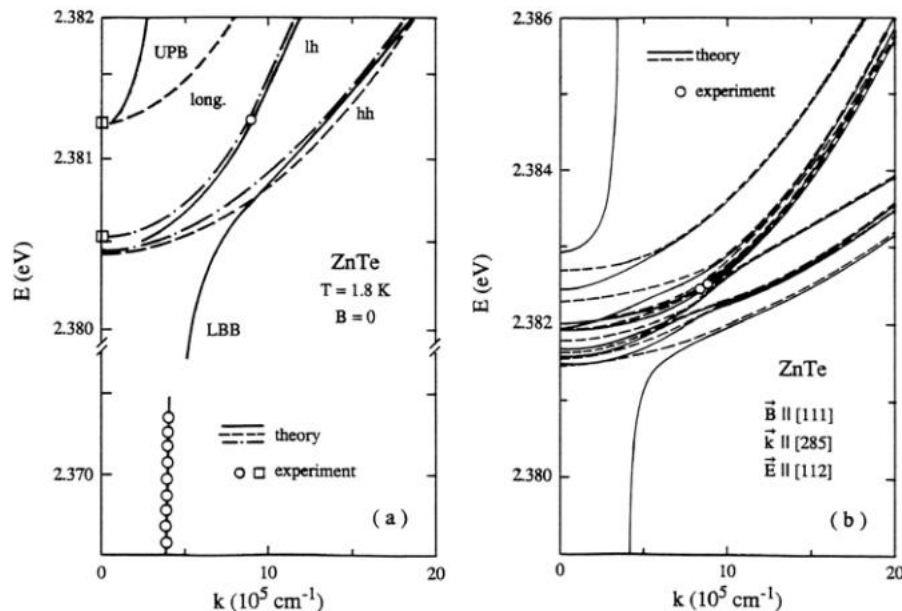


Fig. 16.4. Dispersion relation of the $n_B = 1$ exciton polariton resonances in ZnTe without (a) and with (b) an applied magnetic field [85H1]

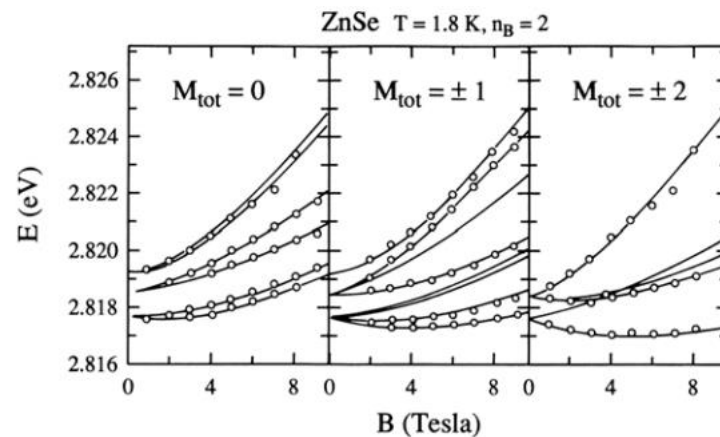


Fig. 16.5. The influence of a magnetic field on the $n_B = 2$ exciton resonances in ZnSe, determined in two-photon absorption spectroscopy [82H1]

- Fig. 16.6 An example the reflectivity of ZnO at the beginning of the continuum states of the C-exciton series.

The Landau-level structure becomes obvious for the highest B-fields.

- In semiconductors with relatively small exciton binding energies like GaAs, the magnetic field may shift states into the continuum, resulting in Fano-interference-type lineshapes.

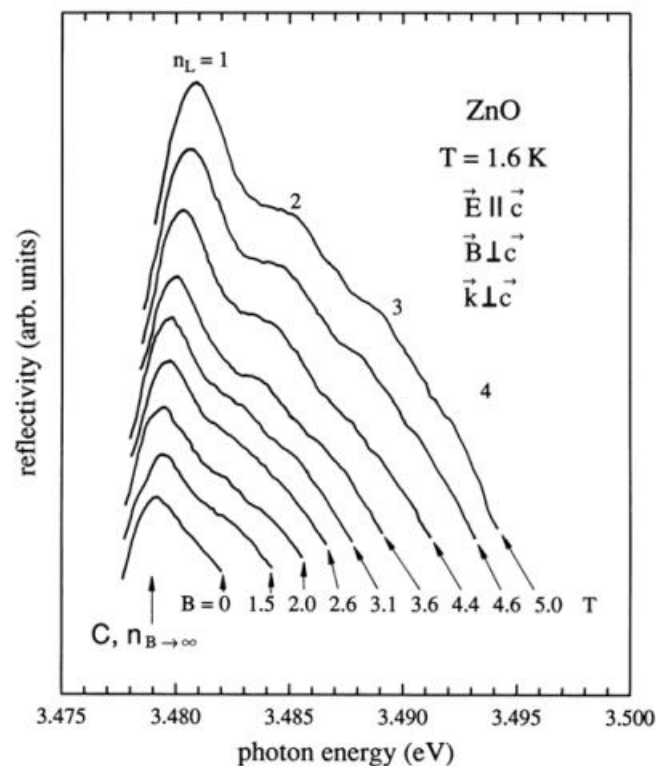


Fig. 16.6. The reflectivity in the exciton continuum in ZnO for various magnetic fields [73H1]

■ 16.1.2 Diluted Magnetic Bulk Semiconductors

Diluted or semimagnetic semiconductors: DMS or SMSC

- The DMS: II–VI 、 III–V semiconductors doped with magnetic ions up to several tens of percent.
- The doping atoms must carry, by definition, spin and magnetic moment.
 - The most widely: Mn^{2+} with a half-filled 3d shell but also the influences of Fe, Co, Ni, V, Cr, Eu or Gd are investigated.
 - The most widely used host materials are narrow and wide gap II–VI materials like $\text{Hg}_{1-x}\text{Mn}_x\text{Te}$, $\text{Cd}_{1-x}\text{Mn}_x\text{Te}$ or $\text{Zn}_{1-x}\text{Mn}_x\text{Se}$.
- Antiferromagnetic ordering: MnTe and $\text{Cd}_{1-x}\text{Mn}_x\text{Te}$ ($x \sim 0.35$).
 - Ferromagnetic like: $\text{Ga}_{1-x}\text{Mn}_x\text{As}$ or $\text{Ga}_{1-x}\text{Mn}_x\text{N}$ or $\text{ZnO}:\text{V}$ and $\text{ZnO}:\text{Co}$.
 - The ferromagnetism can be influenced by the concentration of free carriers.
 - At low concentration they are paramagnetic with very high saturation magnetization and can exhibit spin glass behavior at low T .

- The specific semimagnetic properties for bulk DMS.
- The exchange interaction occurs between the s- and/or p-type conduction and valence bands of the host and the 3d orbitals of the magnetic ions.(Fig. 16.7)
- Free and even more bound carriers can form so-called magnetic polarons.

The formation time is in the range of 100 ps or longer.

The formation of the magnetic polaron cloud lower the energy of the carrier.

This energy decreases with increasing temperature and increasing external magnetic field.

- In narrow gap DMS one observes with increasing magnetic field well-resolved Landau-level fans.

Mn 2+ ions in II–VI DMS show an internal transition resulting in a luminescence band around 2.2 eV.

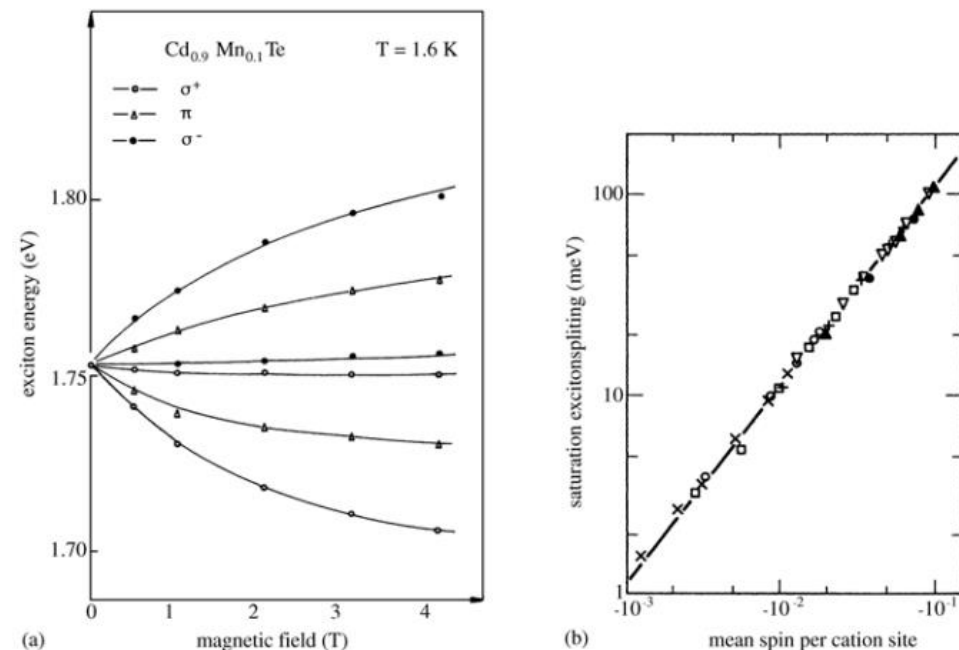
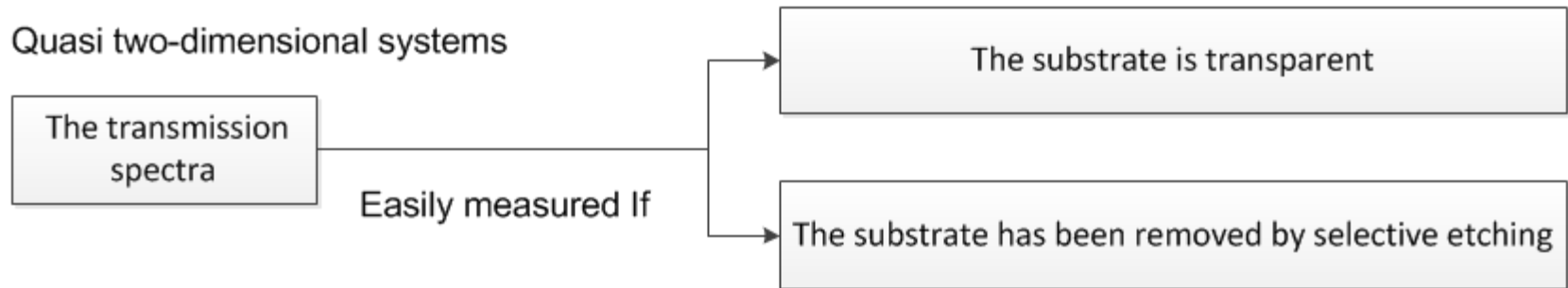


Fig. 16.7. (a) Zeeman splitting of the exciton of $\text{Cd}_{0.9}\text{Mn}_{0.1}\text{Te}$ as a function of magnetic field in the Faraday configuration, i.e., $\mathbf{k} \parallel \mathbf{B}$. (b) Saturation Zeeman splitting per average Mn^{2+} spin per cation site [88G1]

■ 16.1.3 Semiconductor Structures of Reduced Dimensionality



- Many properties are similar to bulk samples.
 - A diamagnetic shift proportional to the area of the exciton in the quantum well if B is normal to it.
 - A Zeeman splitting.
 - There are Landau levels.
- The electron and hole states are fully quantized in all three directions for a magnetic field normal to the quantum well.
- Fig. 16.8 An example for the Landau-levels of a GaAs quantum well sample with 8.5 nm-wide wells for magnetic fields up to 12 T.

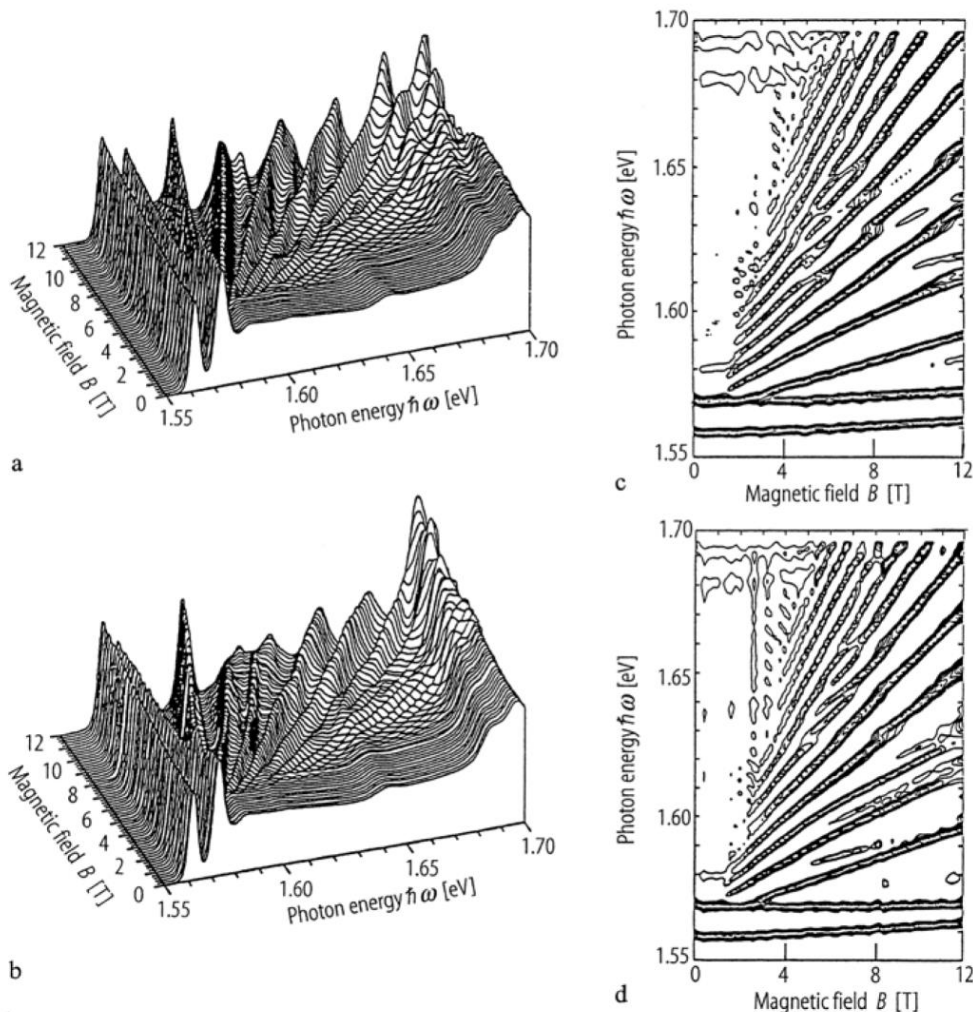


Fig. 16.8. The linear magneto absorption spectra for two different circular polarizations of the light (a, b) The resulting contour plots (c, d) [91S1,92S1]

The two features at lowest energy, the heavy and light hole excitons, are only weakly affected by $\mathbf{B}$ but energetical above them beautifully develops the fan of Landau levels.

- New phenomena:
 - Due to the strong anisotropies of quantum wells and wires, the magnetic field effects depend not only on its orientation to the crystallographic axes but also on its orientation relative to the well or wire.
 - With decreasing well width the wave function penetrates deeper into the barrier. (Fig.16.9) Consequently influence of the magnetic properties of the barrier.
- In diluted magnetic semiconductors the magnetic ions can be incorporated into one or both barriers or into the well itself, allowing information on the interface to be obtained.

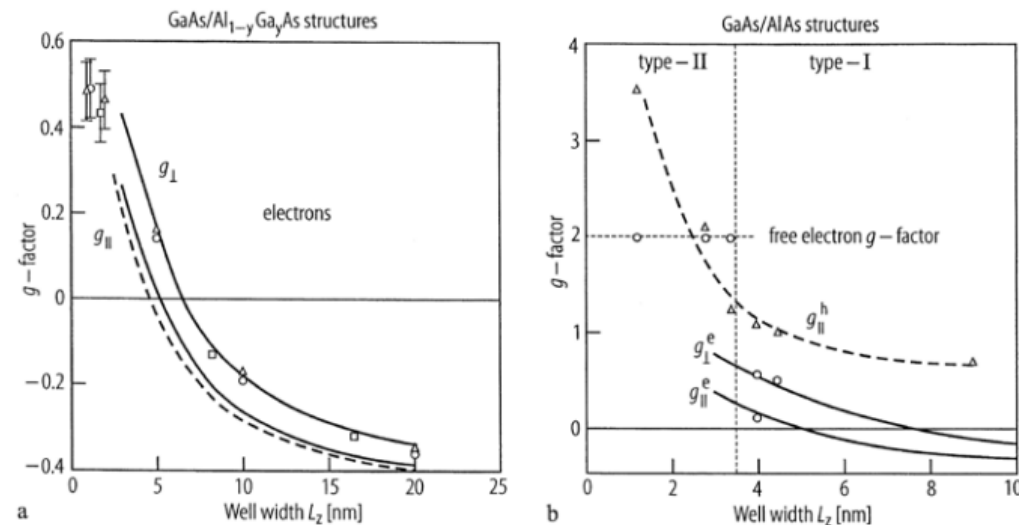


Fig. 16.9. Dependence of electron (and hole) g -factors on the well width and on the orientation of $\mathbf{B}$ in GaAs/Al_{1-y}Ga_yAs (a) and GaAs/AlAs structures (b). In (b) the transition from a type I to a type II band alignment is also indicated (see Sect. 15.2) [92I1,95H1]

16.2 The Effects of Electric Field

If apply a static electric field $\mathbf{E}_s$ on the exciton resonances in semiconductors:

- ❑ There will be a Stark effect.
- ❑ The influence of E_s on the band-to-band transitions, on the continuum states is also known as the Franz–Keldysh effect- modification on the absorption spectrum - resulting in a red-shift of the band-to-band absorption tail
- ❑ Influence the refractive index in the transparent spectral region below the exciton resonances, resulting in Pockels- or Kerr-effect-like phenomena.
- ❑ The influence of $\mathbf{E}_s$ on the eigenfrequencies.

16.2.1 Bulk Semiconductors

It is not easy to observe the Stark effect for excitons in three-dimensional. To get an observable shift of the eigenstates, the electric field is in the order of 10^6 Vm^{-1}

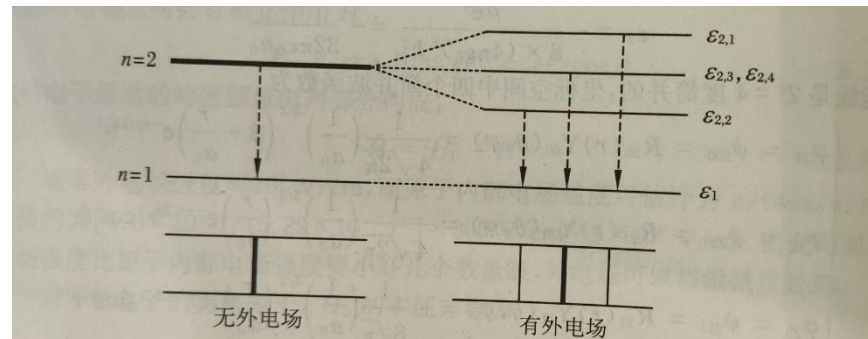
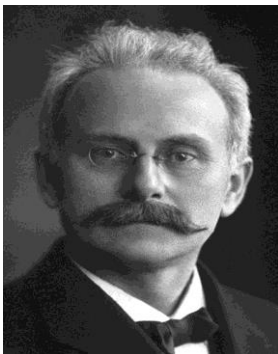
Such fields broaden or even destroy the exciton resonances due to two effects.

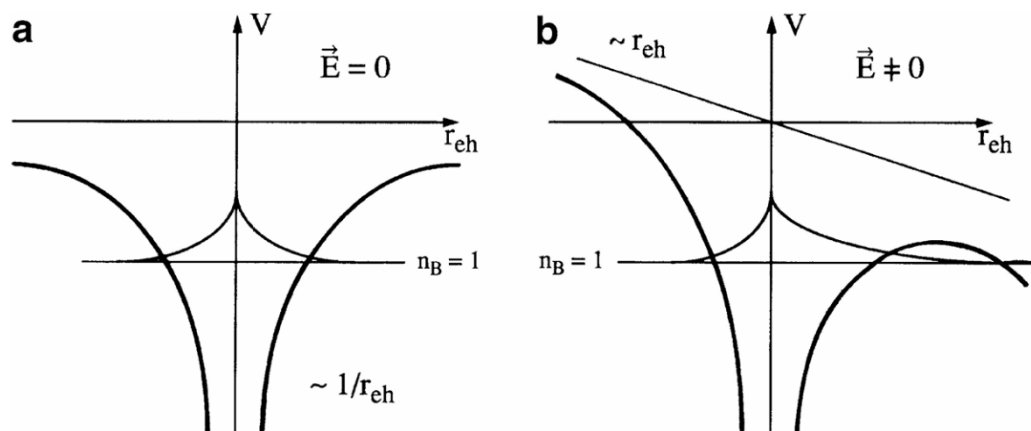
- One is field ionization of the exciton by tunneling through the finite Coulomb barrier at finite fields (Fig.16.10)
- The other is impact ionization.

Stark Effect

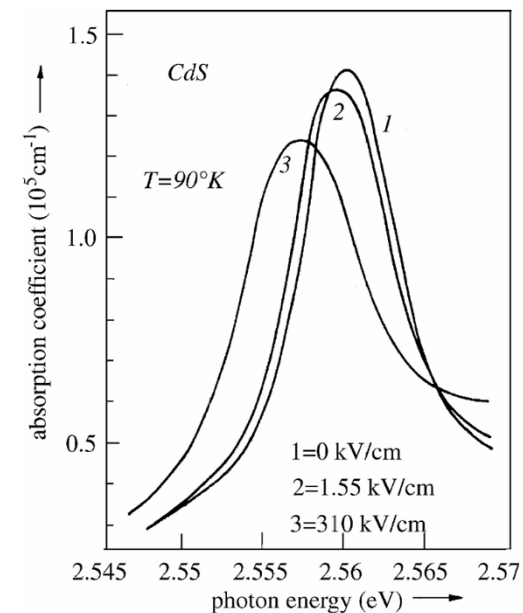
The Stark Effect refers to the splitting of spectral lines of atoms when exposed to an external static electric field, arising from the perturbation of atomic energy levels by the electric field

- First observed in 1913 by the German physicist Johannes Stark, who detected spectral line splitting while studying the hydrogen atom spectrum
- Similar to the Zeeman Effect (spectral splitting caused by magnetic fields), the Stark Effect specifically resulting from electric field-induced changes to atomic energy levels
- Atomic electron energy levels initially exhibit "degeneracy"—different quantum states correspond to the same energy value
- An external static electric field breaks this degeneracy, splitting originally overlapping energy levels into multiple sub-levels with distinct energies
- Linear Stark Effect: The degree of splitting is proportional to the strength of the external electric field. It is commonly observed in low energy levels of simple systems like hydrogen atoms
- Quadratic Stark Effect: The splitting degree is proportional to the square of the electric field strength. It typically occurs in strong electric field environments or complex atomic systems





The Coulomb potential of a three-dimensional exciton without (a) and with (b) an applied constant electric field

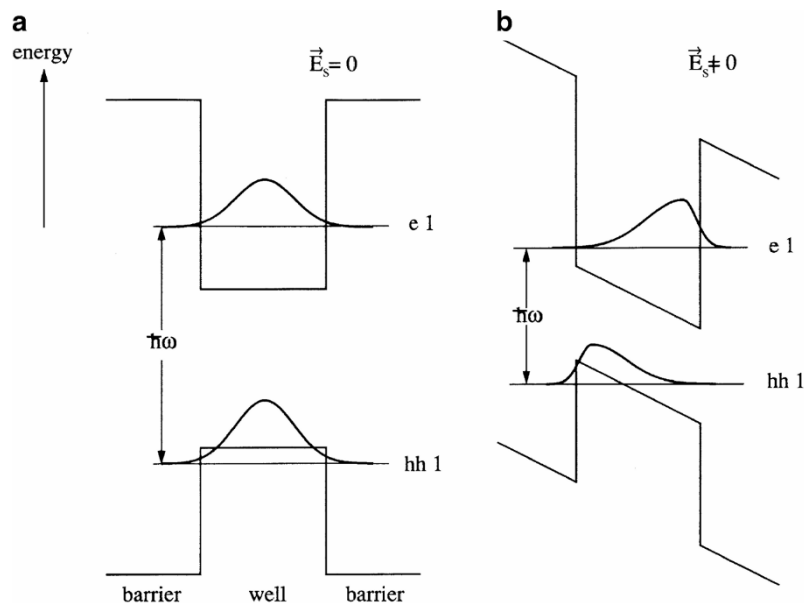


Influence of an external field on the absorption spectrum of the B, $n_B = 1$ exciton of CdS for various applied electric fields

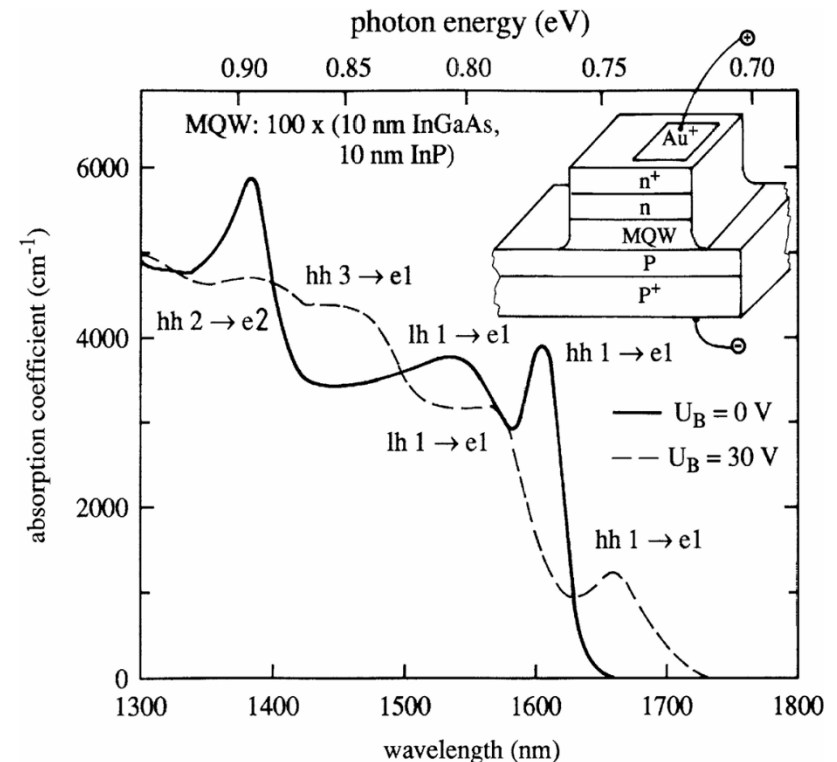
16.2.2 Stark Effect in Semiconductor Structures of Reduced Dimensionality-QCSE

By confining the electron and hole between barriers, e.g., in quantum wells, wires or dots can overcome the problem of field ionisation in bulk semiconductors. Three effects:

- The red-shift of the exciton.
- The decrease of its oscillator strength.
- The appearance of forbidden transitions.

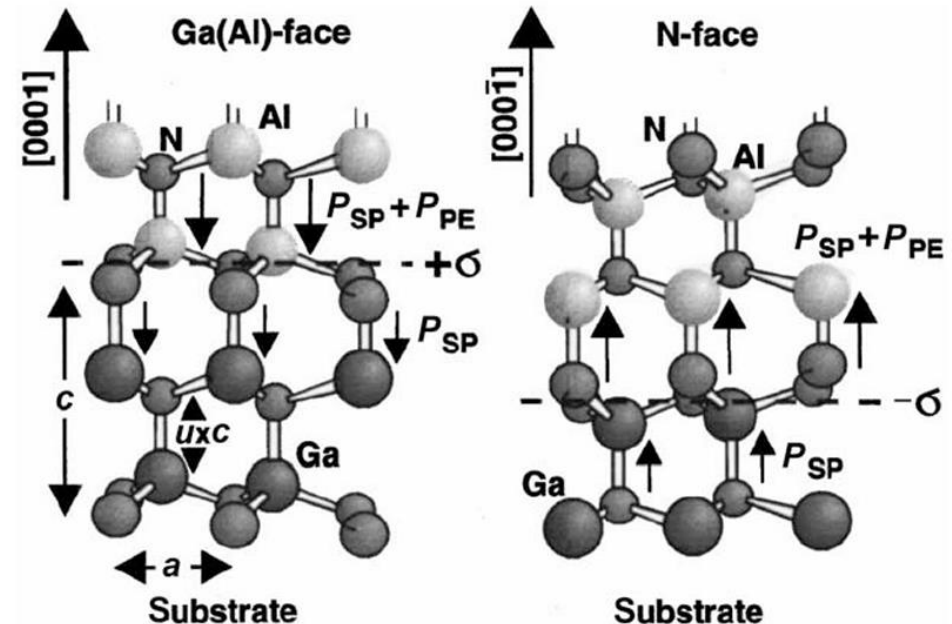
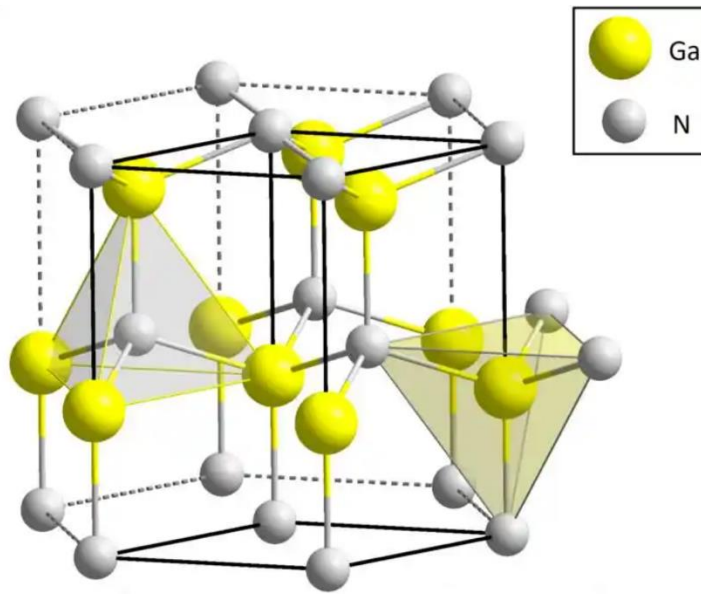


The bandstructure of a quantum well without (a) and with (b) an applied constant electric field perpendicular to the wells, but neglecting the Coulomb interaction between electrons and holes



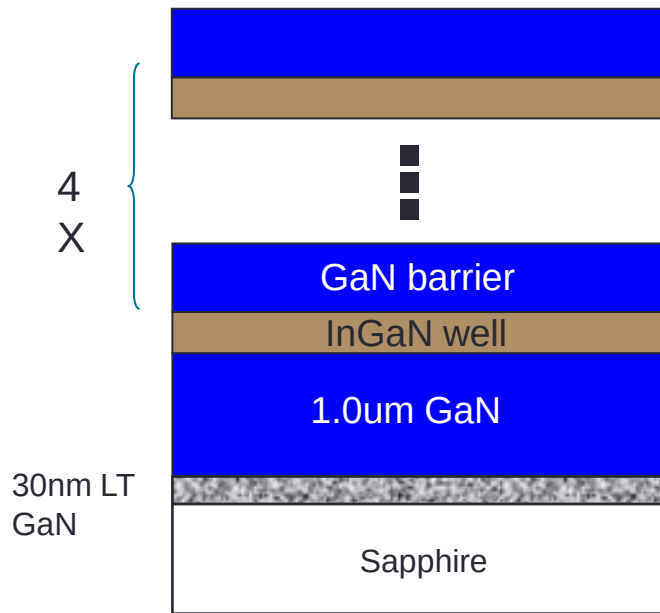
The quantum-confined Stark effect in InGaAs/InP MQW samples

Piezoelectric (PE) and spontaneous polarization (SP) effects in GaN



Piezoelectric (PE) and spontaneous polarization (SP) effects in Ga(Al) or N face AlGa_xN/GaN heterostructures

QCSE in InGaN/GaN QWs

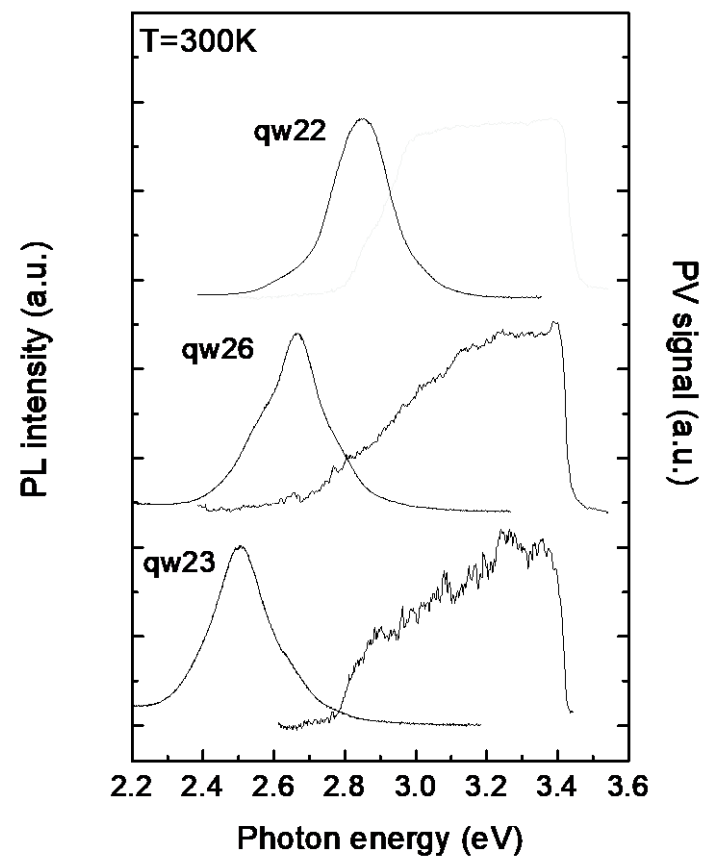
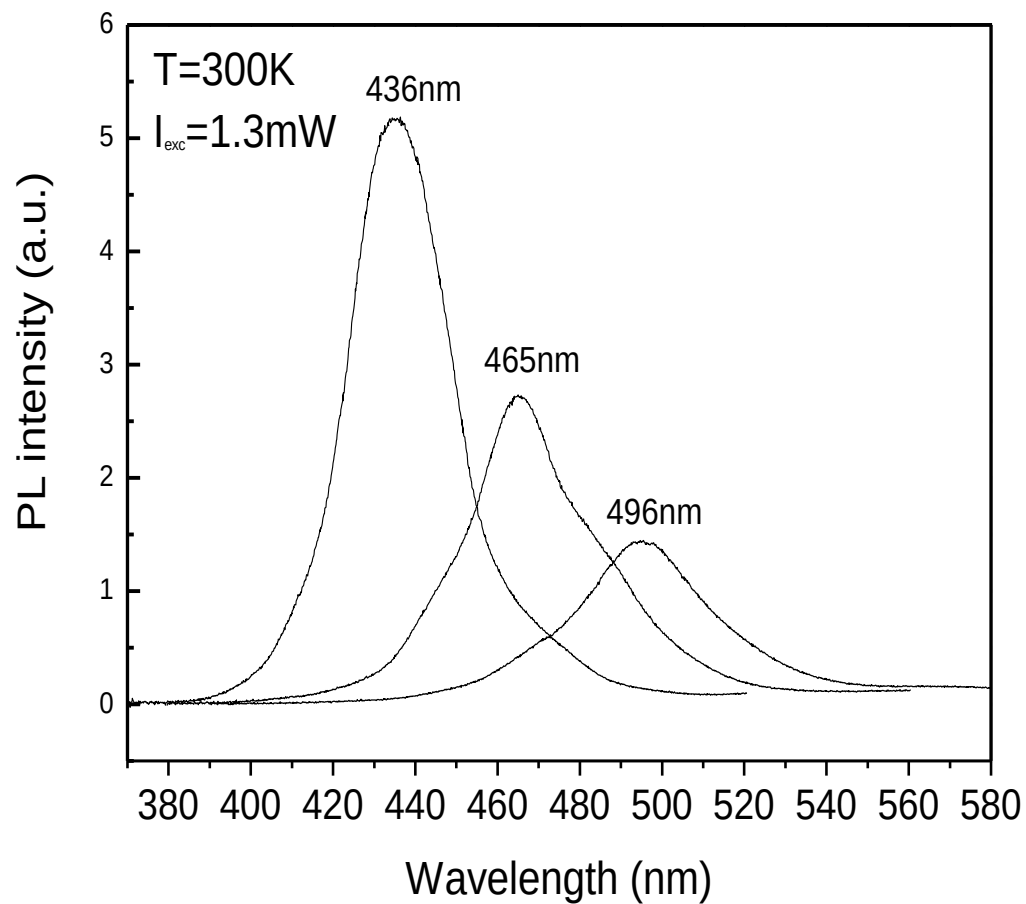


Schematic diagram of sample structure

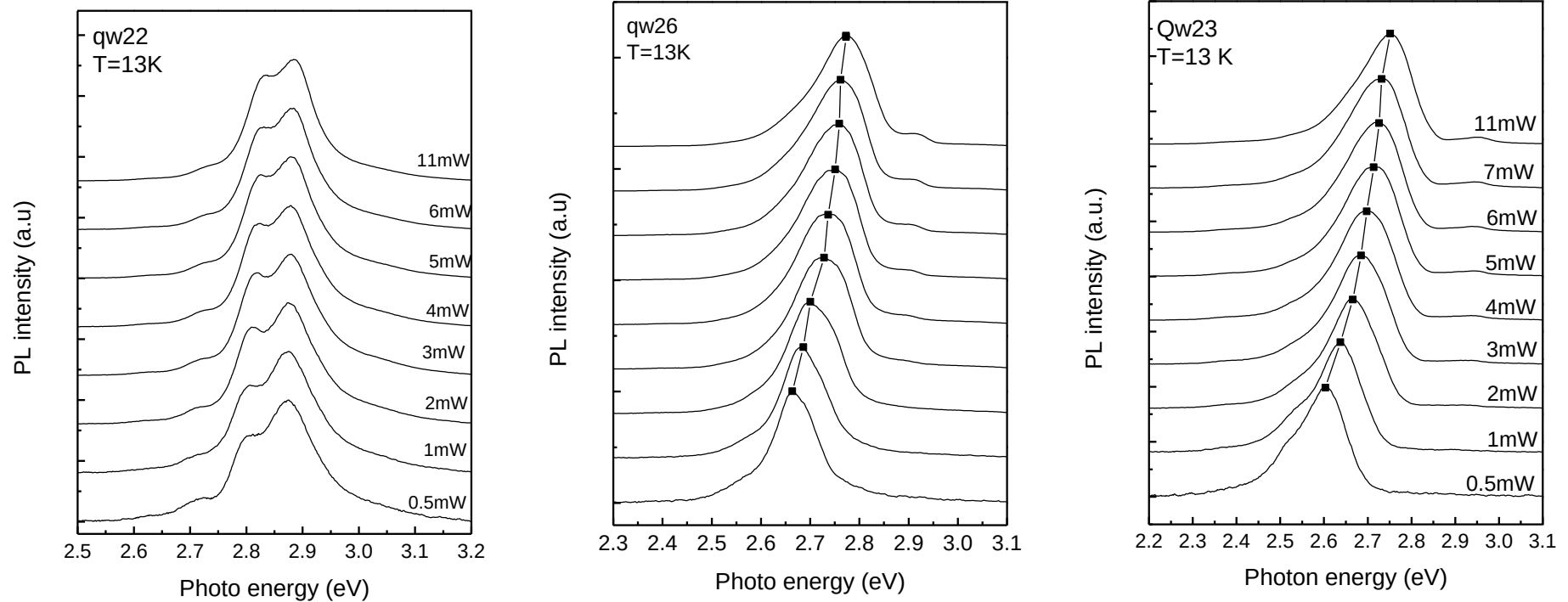
Sample Parameters

Sample	Well thickness (Å)	Barrier thickness (Å)	Indium fraction
qw22	22	176	0.19
qw26	33	176	0.19
qw23	44	176	0.19
qw29	44	176	0.13
qw27	44	176	0.09

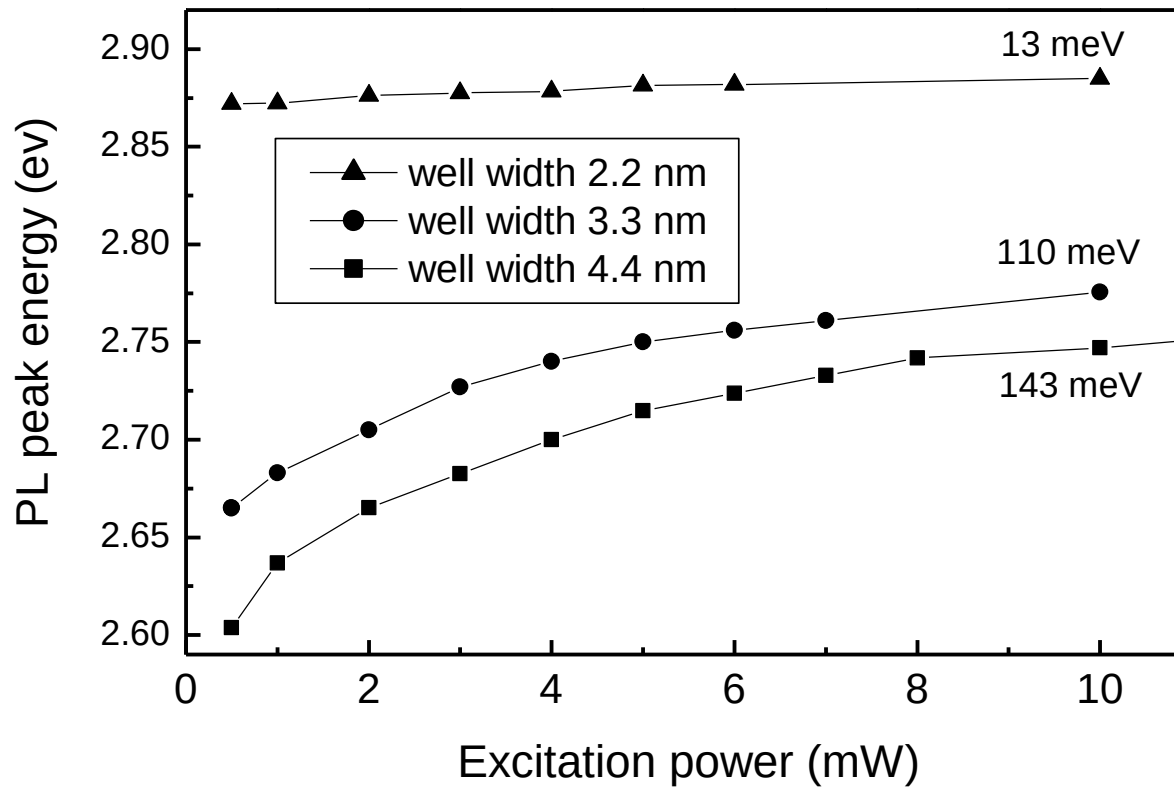
PL and PV Spectra measurement



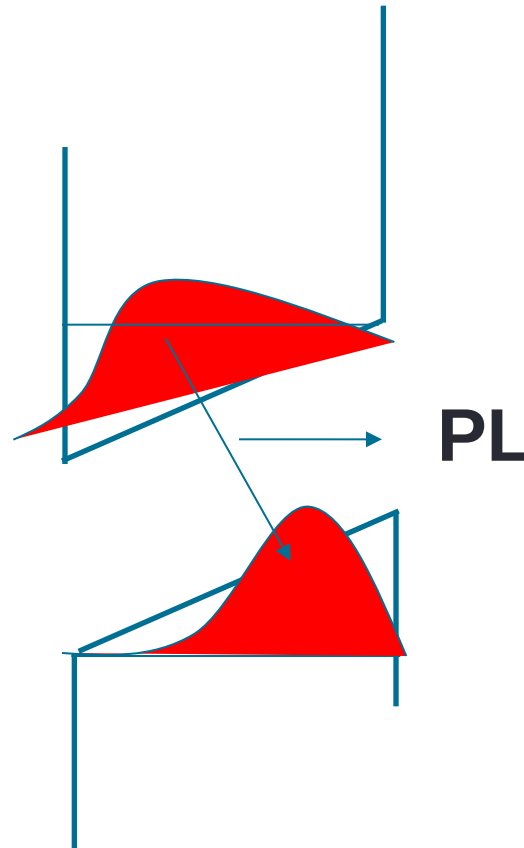
PL Spectra-Excitation Intensity Dependence



Excitation Intensity Dependence

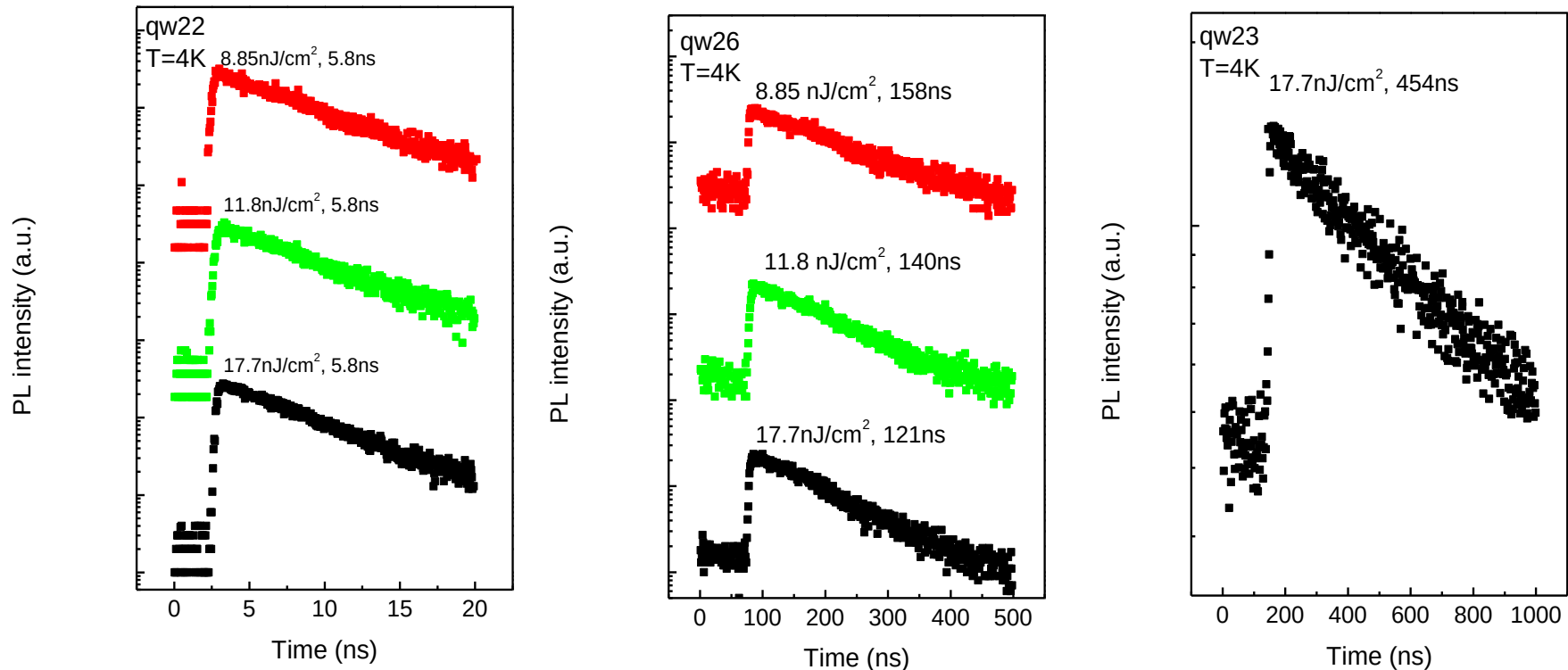


Quantum Confined Stark Effect



Schematic view of optical transitions in the tilt InGaN quantum wells

PL Decay - Well Width Dependence



- The built-in electric fields separate the electrons and holes spatially, reduce the oscillator strength and thus recombination rate, therefore long decay time is anticipated
 - The PL decay time will increase with the well width
 - The PL decay time will be excitation intensity dependent

16.3 Strain Fields on the

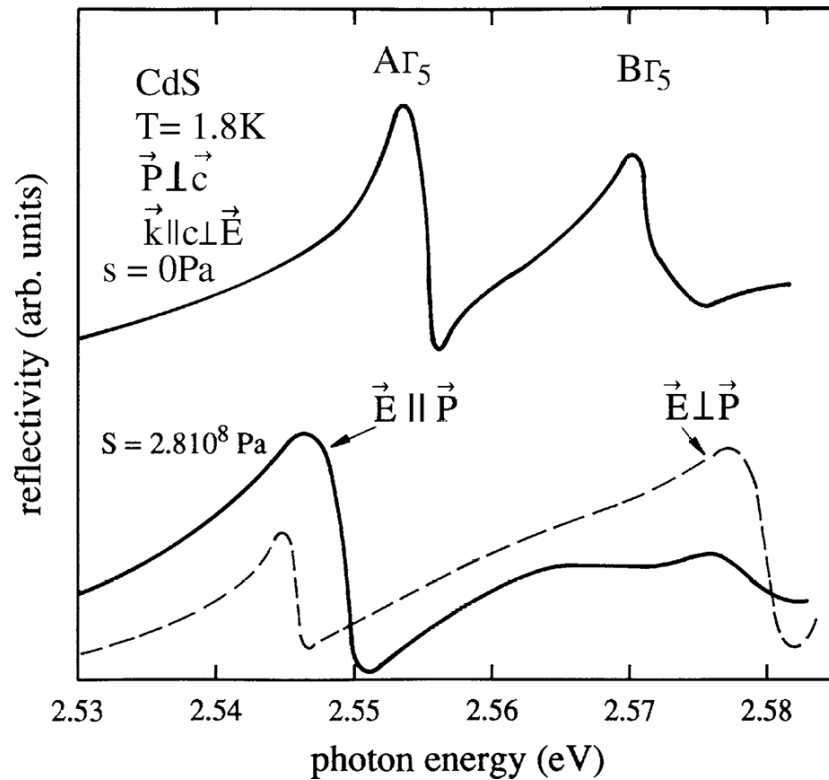
- The dependence of the energy of the band extrema on the strain, the so-called deformation potential is defined for conduction and valence band by

$$\mathcal{E} = a \frac{dE_{c,v}}{dx}$$

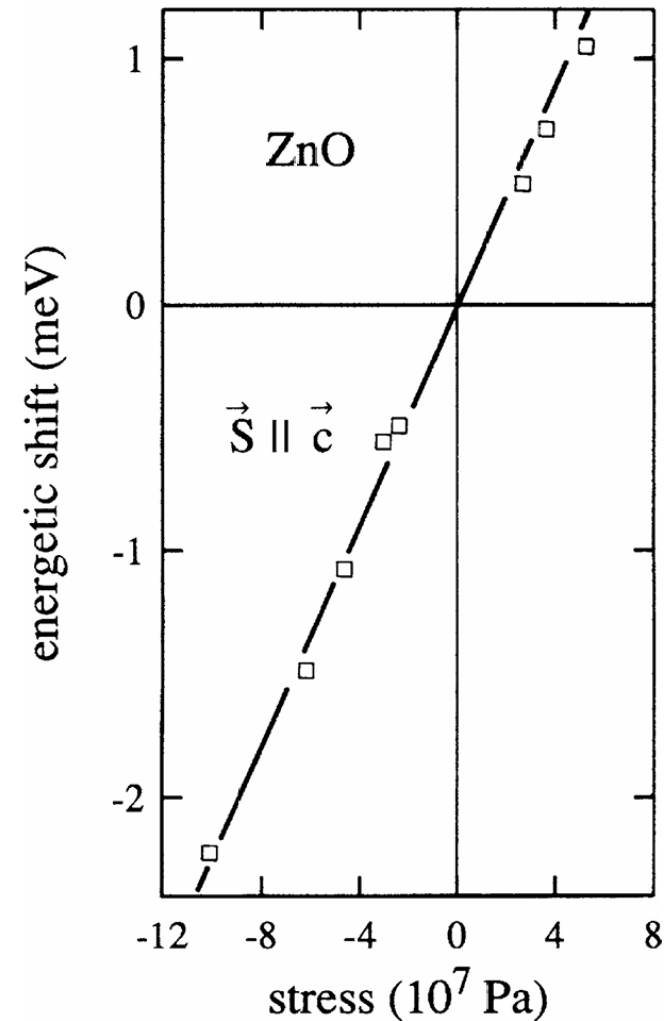
where a is the lattice constant, $dE_{c,v}$ is the shift of the band edge caused by a length variation of the unit cell by an amount dx . Deformation potentials are defined for various types of strain like hydrostatic, uniaxial or shear strain

- For excitons the relevant quantity is the sum effects of the deformation potentials on conduction and valence bands. As in the case of electric field, an applied stress changes the eigenenergies of the exciton states and may also lead to a splitting of degenerate states,

16.3.1 Bulk Semiconductors



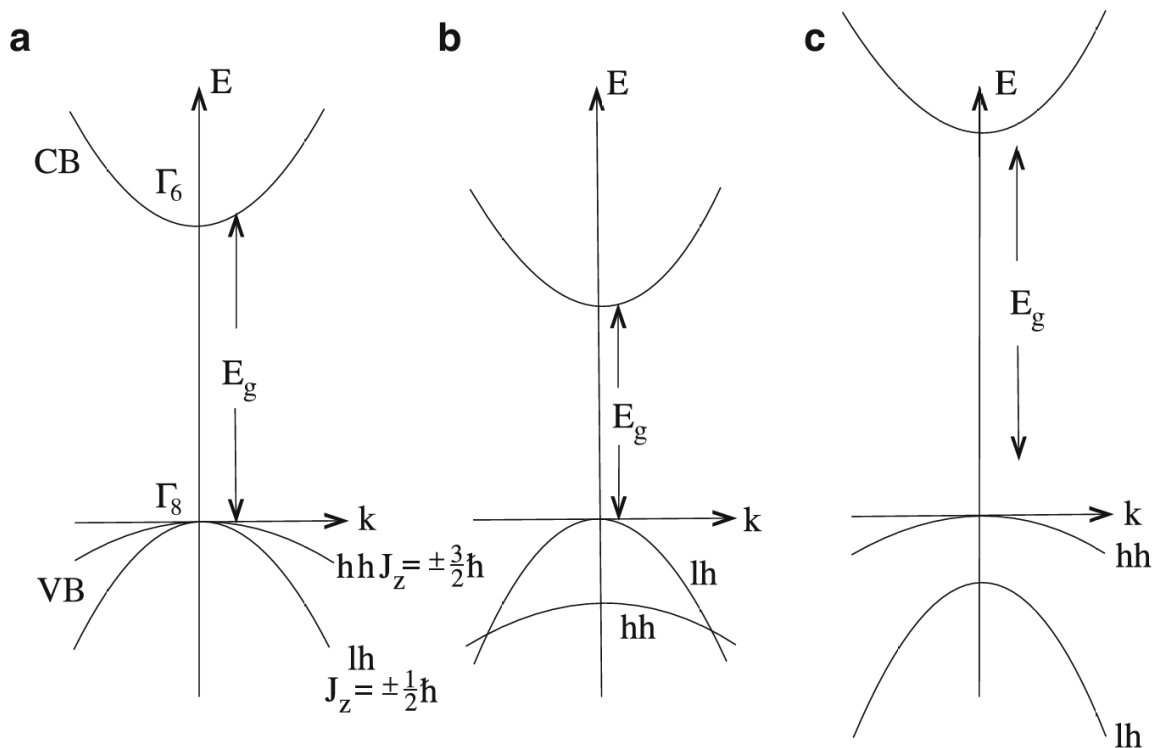
The excitonic reflection spectra of CdS under uniaxial stress



The shift of a BEC luminescence line in ZnO under the influence of strain

16.3 Strain Fields on the Structures of Reduced Dimensionality

Strain-induced splitting of the Γ_8 valence band is of special importance for strained quantum wells and superlattices whenever a layer is grown with a lattice constant different from that of the substrate.



Schematic of the influence of strain on the alignment of the hh and lh valence bands and on the value of the band gap (a) zero strain (b) two-dimensional tensile strain and normal uniaxial compressive strain (c) and two-dimensional compressive strain and normal uniaxial tensile strain.